SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Data Warehousing and Business Intelligence

IT3021 Assignment 2

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1. Data Source Description

The data warehouse used in this project is designed to store and analyze transactional data efficiently. It consists of multiple dimension tables and one fact table that support analytical operations.

The main tables used in this data warehouse include:

Dimensions:

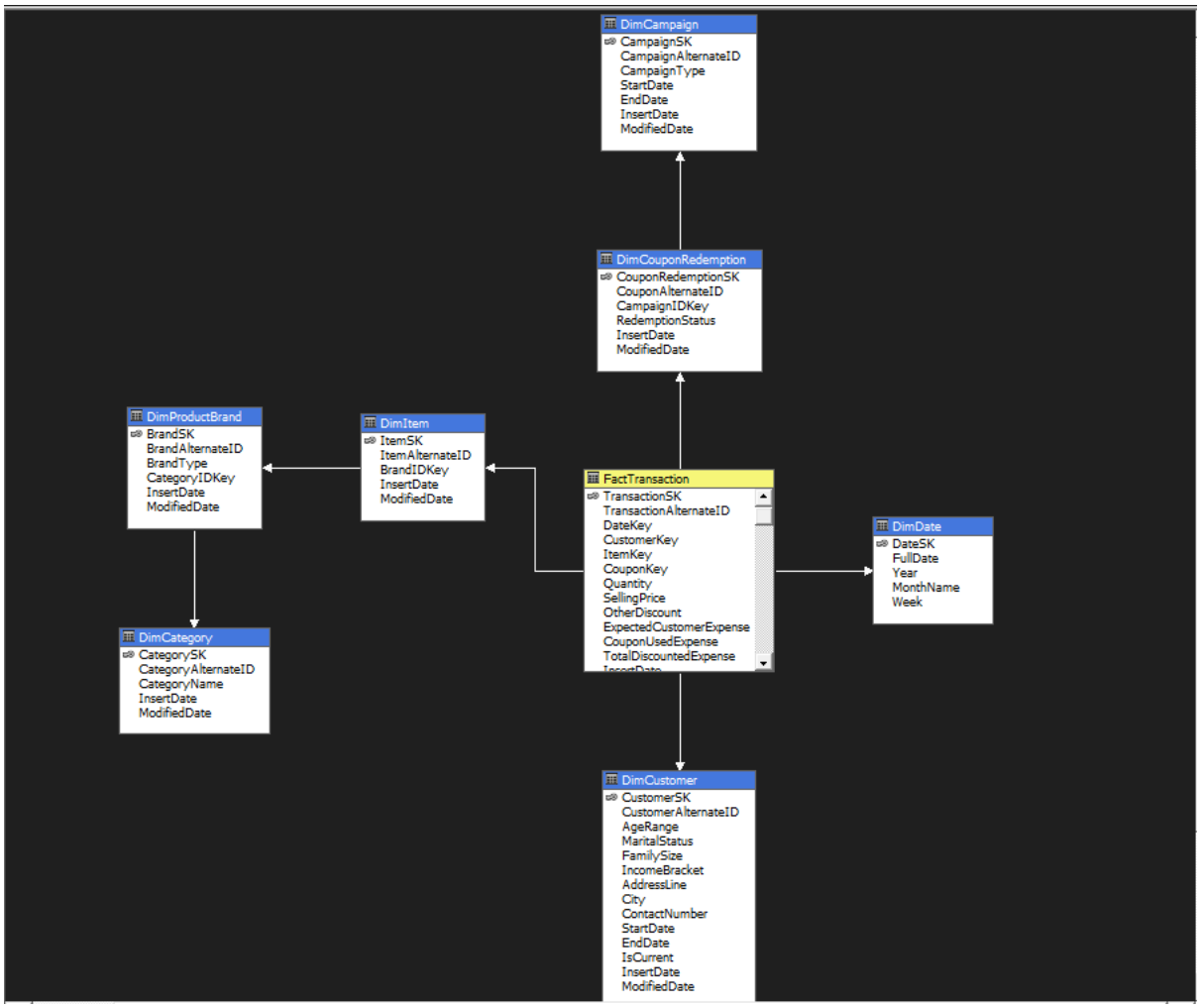
- DimDate
- DimCustomer
- DimItem
- Dim Coupon Redemption
- Dim Category
- Dim Campaign
- Dim ProductBrand

Fact Table:

- FactTransaction

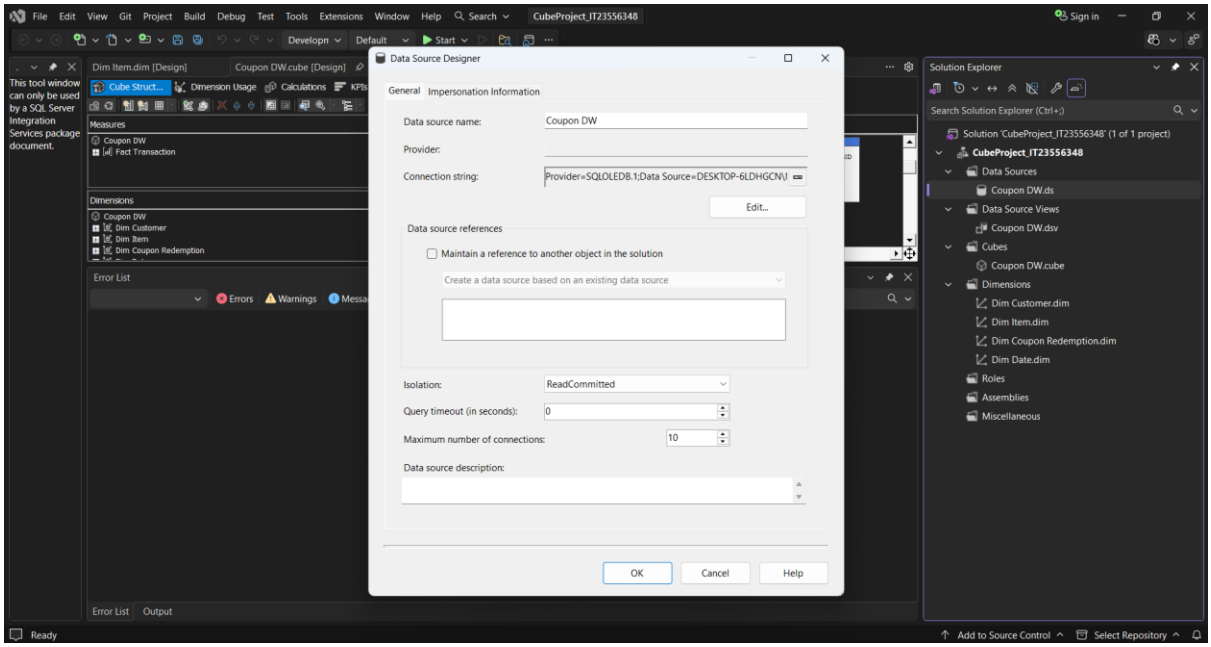
The fact table stores measurable data such as Total Discounted Expense, while dimension tables provide descriptive attributes like date, customer, and item details.

This structure enables efficient querying and supports analytical operations such as slicing, dicing, and drill-down.


DimCampaign
CampaignSK
CampaignAlternateID
CampaignType
StartDate
EndDate
InsertDate
ModifiedDate
DimCouponRedemption
CouponRedemptionSK
CouponAlternateID
CampaignIDKey
RedemptionStatus
InsertDate
ModifiedDate
DimProductBrand
BrandSK
BrandAlternateID
BrandType
CategoryIDKey
InsertDate
ModifiedDate
DimItem
ItemSK
ItemAlternateID
BrandIDKey
InsertDate
ModifiedDate
FactTransaction
TransactionSK
TransactionAlternateID
DateKey
CustomerKey
ItemKey
CouponKey
Quantity
SellingPrice
OtherDiscount
ExpectedCustomerExpense
CouponUsedExpense
TotalDiscountedExpense
DimDate
DateSK
FullDate
Year
MonthName
Week
DimCategory
CategorySK
CategoryAlternateID
CategoryName
InsertDate
ModifiedDate
DimCustomer
CustomerSK
CustomerAlternateID
AgeRange
MaritalStatus
FamilySize
IncomeBracket
AddressLine
City
ContactNumber
StartDate
EndDate
IsCurrent
InsertDate
ModifiedDate

2. Cube Creation and Deployment

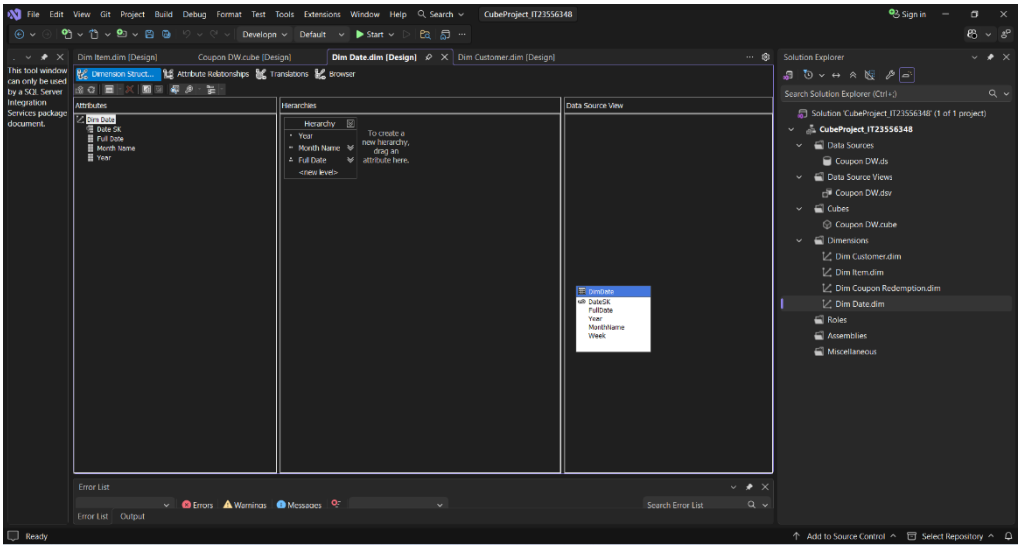
After designing the data warehouse, an **SSAS (SQL Server Analysis Services)** cube. The cube was designed using the data warehouse as the data source. In the cube, the fact table was linked to the dimensions and various measures were incorporated in to cube for analysis.

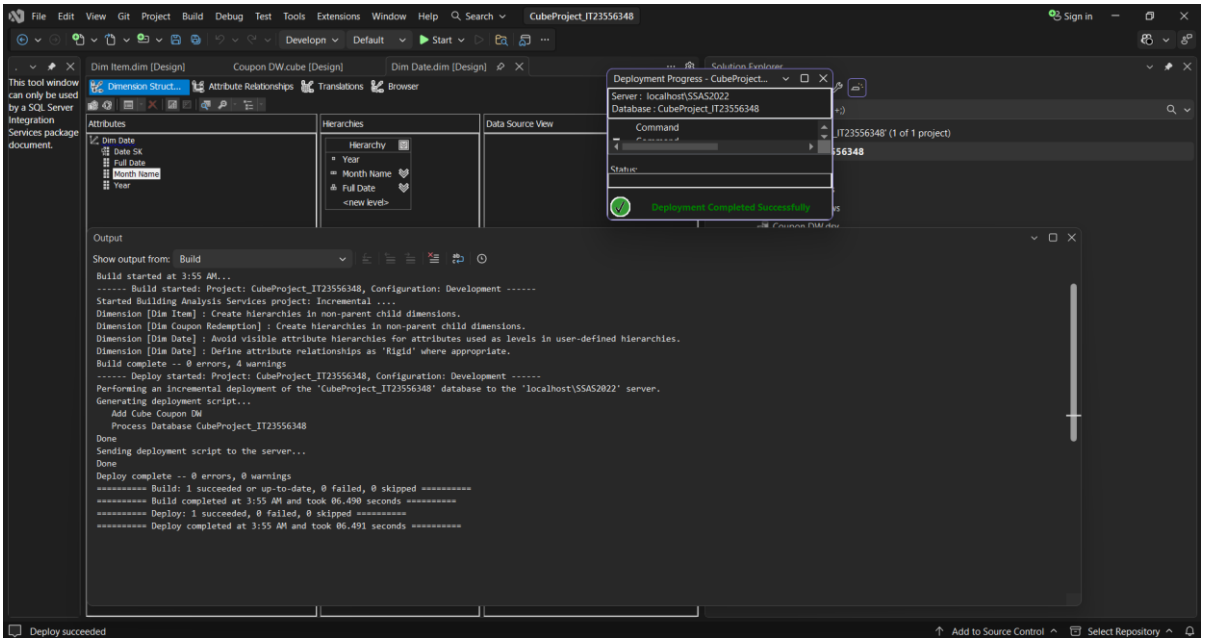


The cube includes:

- **Measures:** Total Discounted Expense
- **Dimensions:** Date, Customer, and Item
- **Hierarchies:** Year → Month → Full Date

The cube was **deployed** successfully, allowing users to perform OLAP operations such as roll-up, drill-down, slice, and dice.

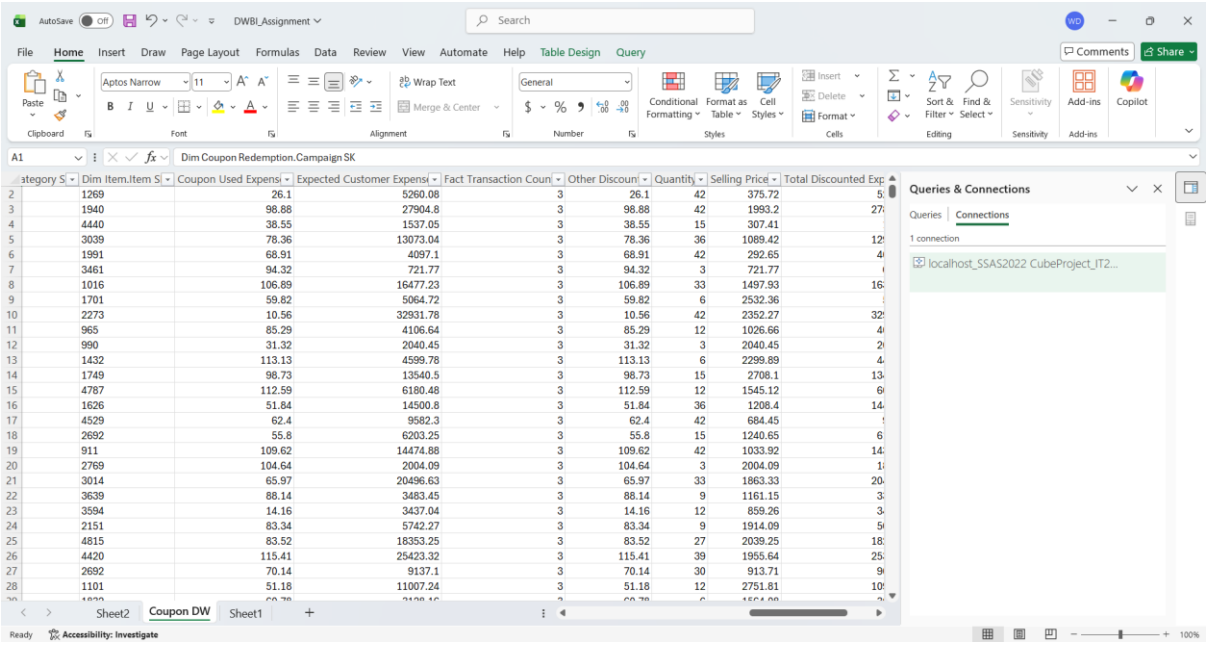




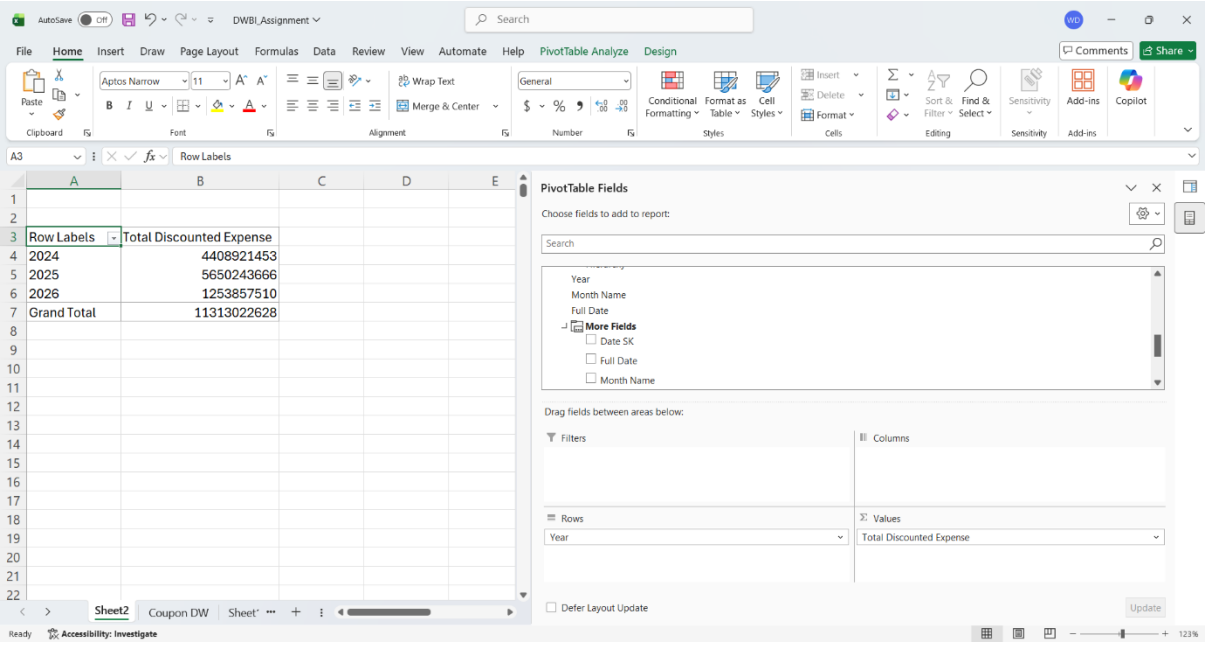
- Although multiple dimension tables are available in the data warehouse, only the relevant dimensions were selected for cube design and reporting. Including all dimensions would increase complexity and reduce performance without adding analytical value. Therefore, only key dimensions such as Date, Customer, Item, and Coupon Redemption were used to support meaningful analysis.

3. OLAP Operations in Excel

1.Connection to SSAS Cube: The SSAS cube was connected to Excel using the Data tab and PowerPivot mode for data analysis.

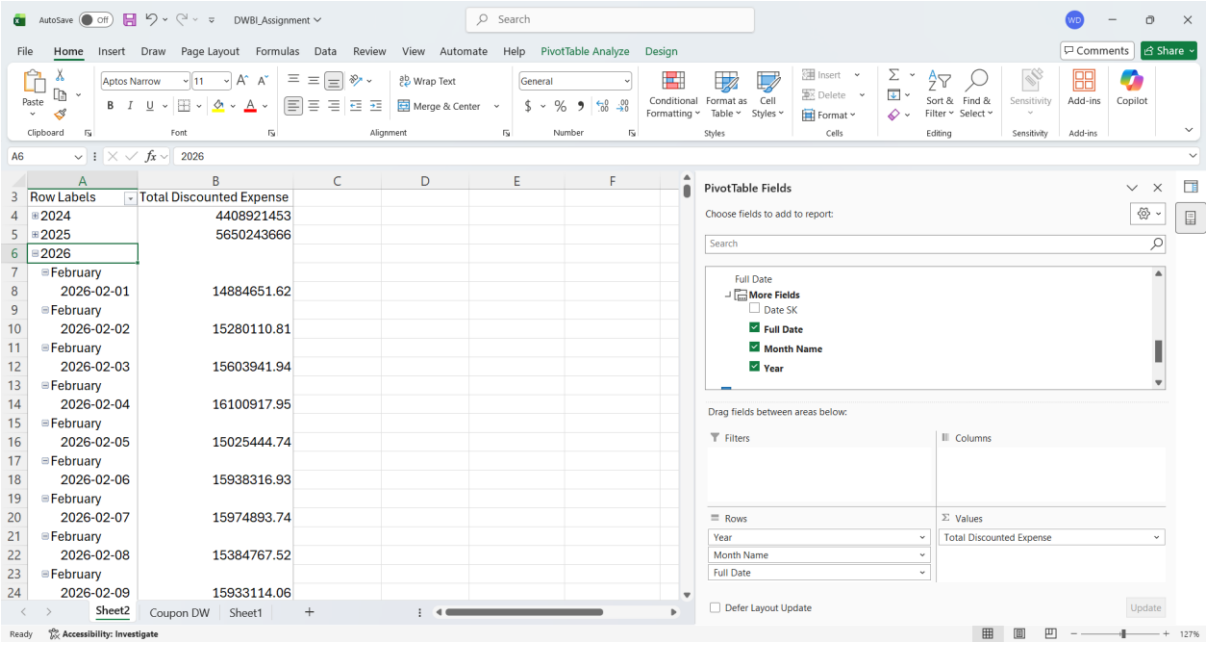


2. Roll up - Data aggregated at yearly level using roll-up operation**.**



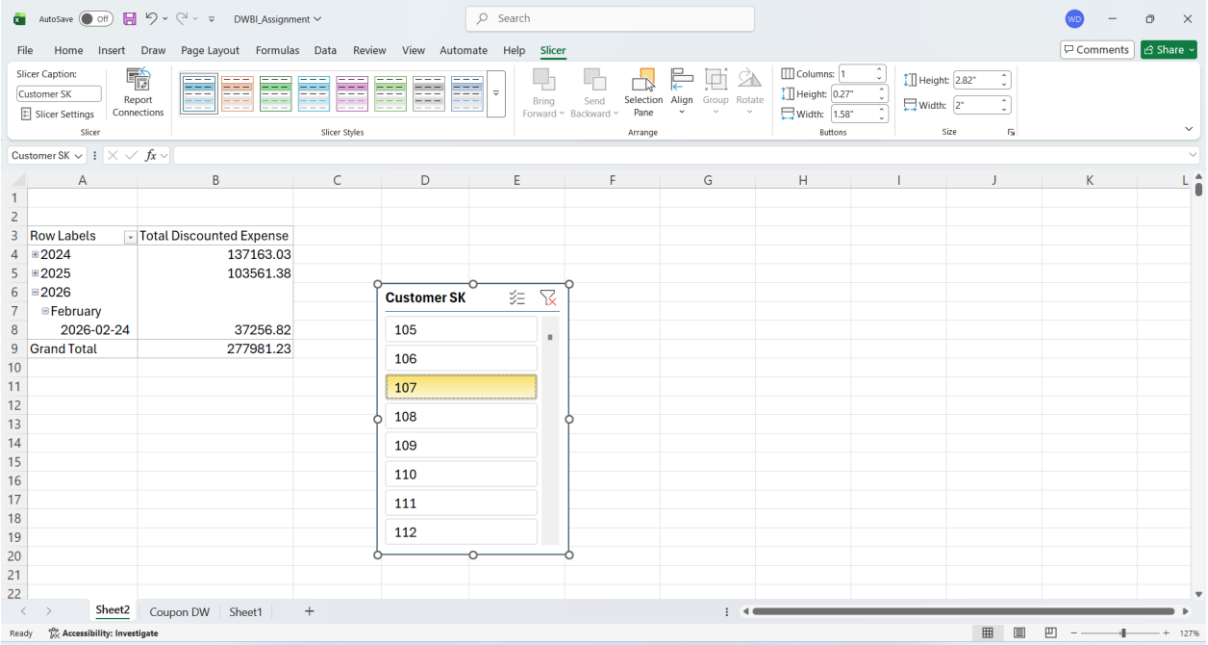
Row Labels	Total Discounted Expense
2024	4408921453
2025	5650243666
2026	1253857510
Grand Total	11313022628

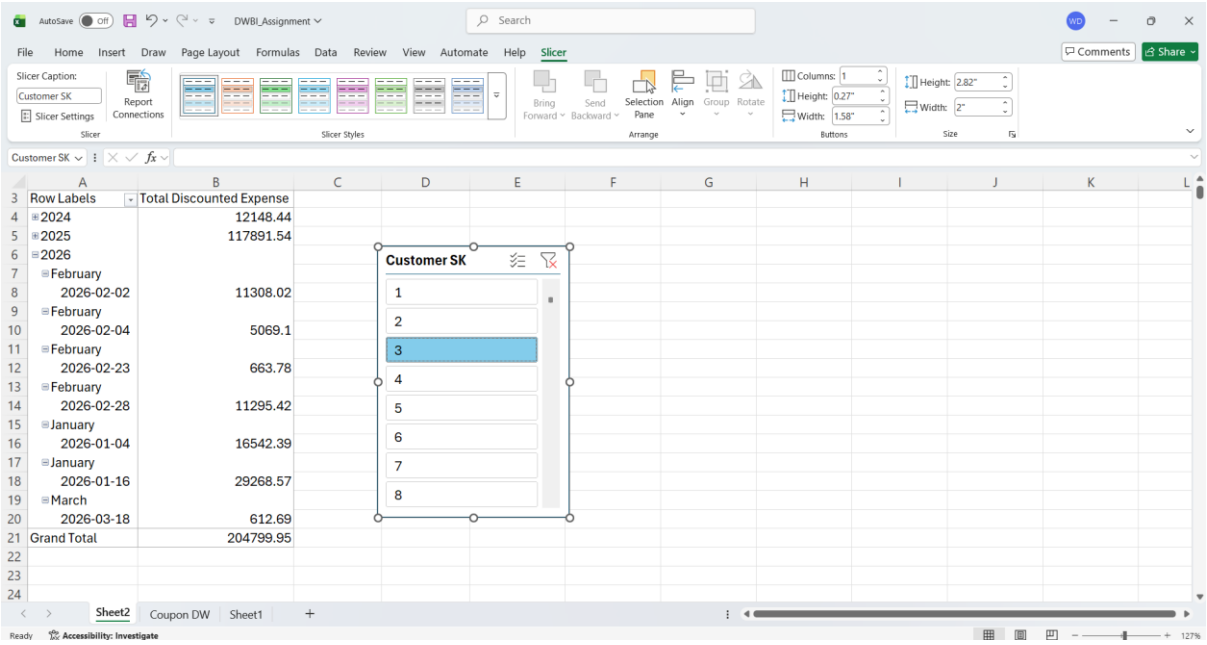
3. Drill Down - Drill-down performed from Year to Month to Date.



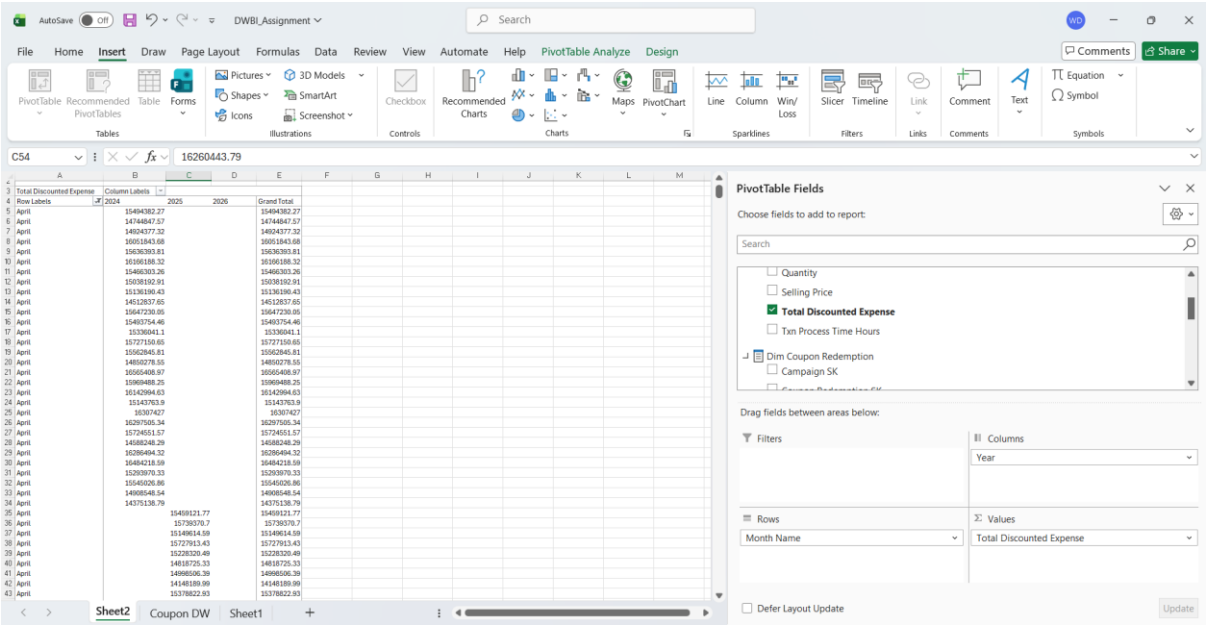
Row Labels	Total Discounted Expense
2024	4408921453
2025	5650243666
2026	
February	
2026-02-01	14884651.62
February	
2026-02-02	15280110.81
February	
2026-02-03	15603941.94
February	
2026-02-04	16100917.95
February	
2026-02-05	15025444.74
February	
2026-02-06	15938316.93
February	
2026-02-07	15974893.74
February	
2026-02-08	15384767.52
February	
2026-02-09	15933114.06

4.Slice - Slice operation performed using slicers (Customer)

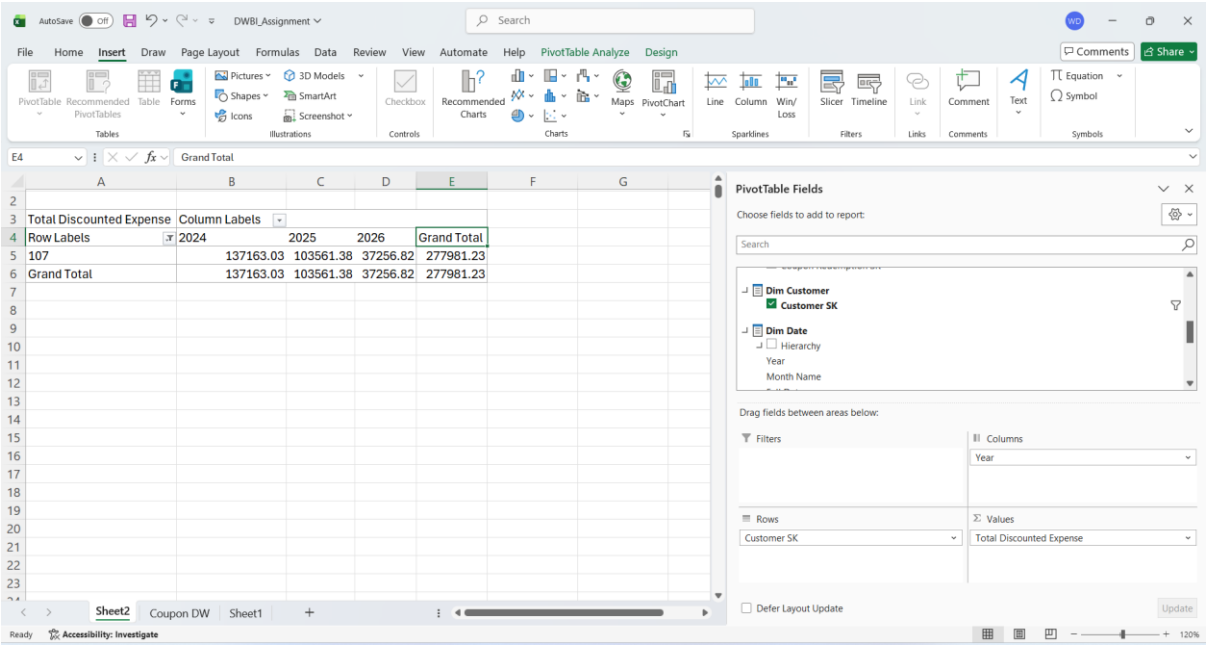




5.Dice – Data analysed across multiple dimensions (Year & Month)



6. Pivot - Data orientation changed for different perspective

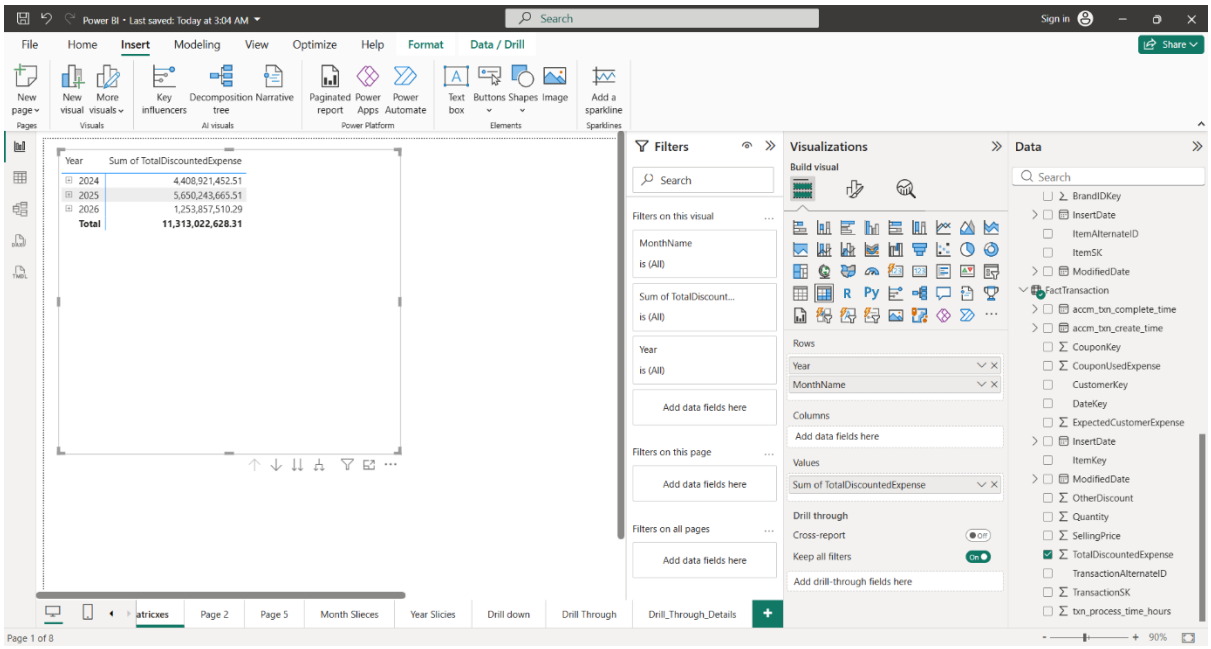


4. Power BI Reports

1. Report 1 (Matrix Visual): The Matrix visual is used to perform roll-up analysis on the data warehouse. In this project, the matrix **summarizes the Total Discounted Expense across different years.**

The rows contain the Year dimension, while the values display the aggregated **TotalDiscountedExpense** measure. This allows users to view high-level summaries such as total expenses for each year (2024, 2025, 2026), as well as the overall grand total.

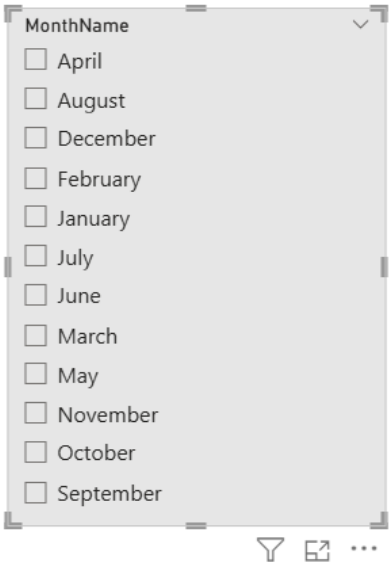
These visual supports hierarchical analysis, where users can expand or collapse levels (e.g., Year → Month) to explore data at different granularities

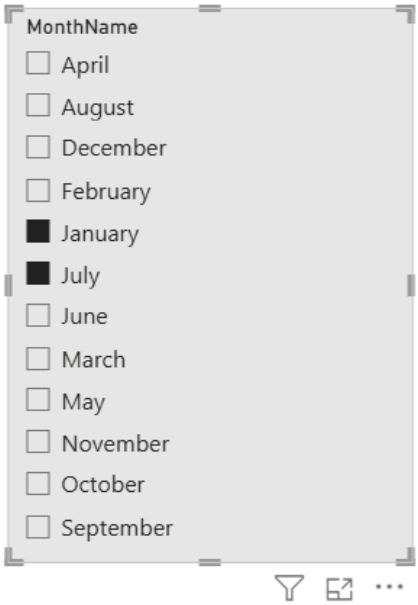


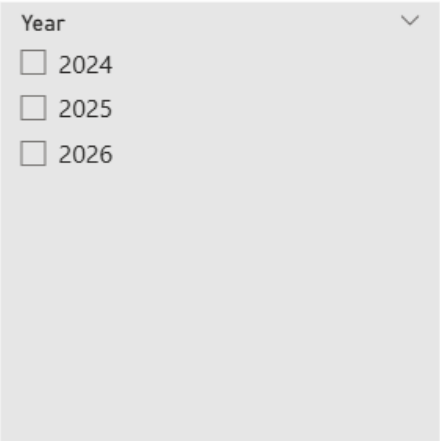
2. Report 2 (Multiple Slicers) : Slicers are used to perform the slice operation in OLAP. In this project, slicers are created for **MonthName and Year dimensions**.

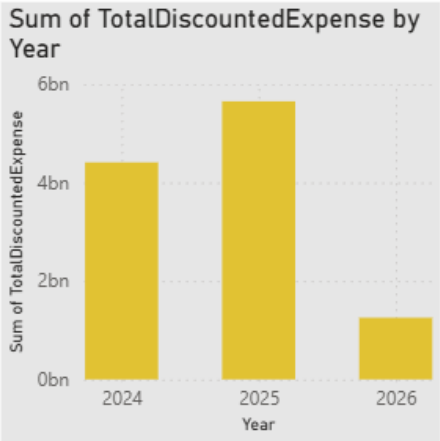
The slicers allow users to filter the report interactively. For example, selecting a specific month (e.g., April) or year (e.g., 2025) updates all related visuals such as charts and matrices instantly.

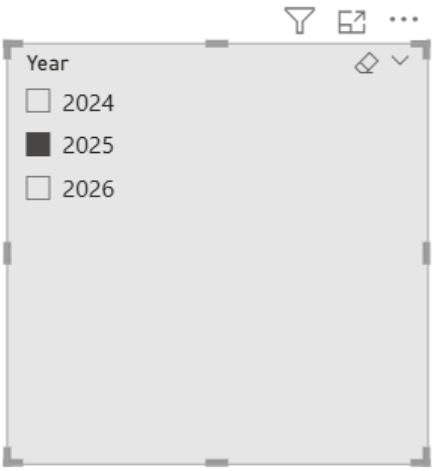
This helps users analyse data for specific conditions without modifying the underlying dataset, improving usability and interactivity of the dashboard

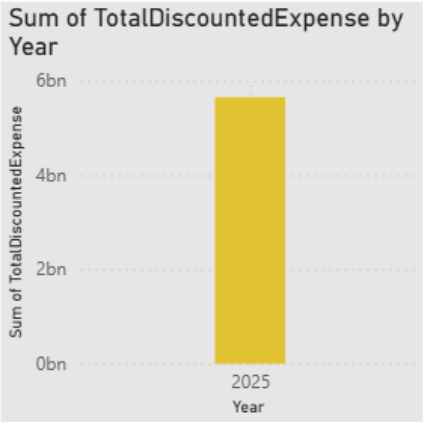








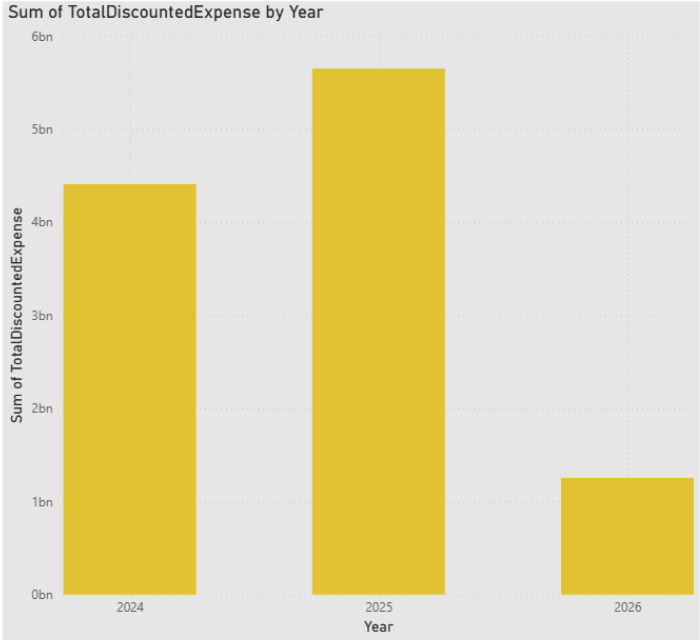




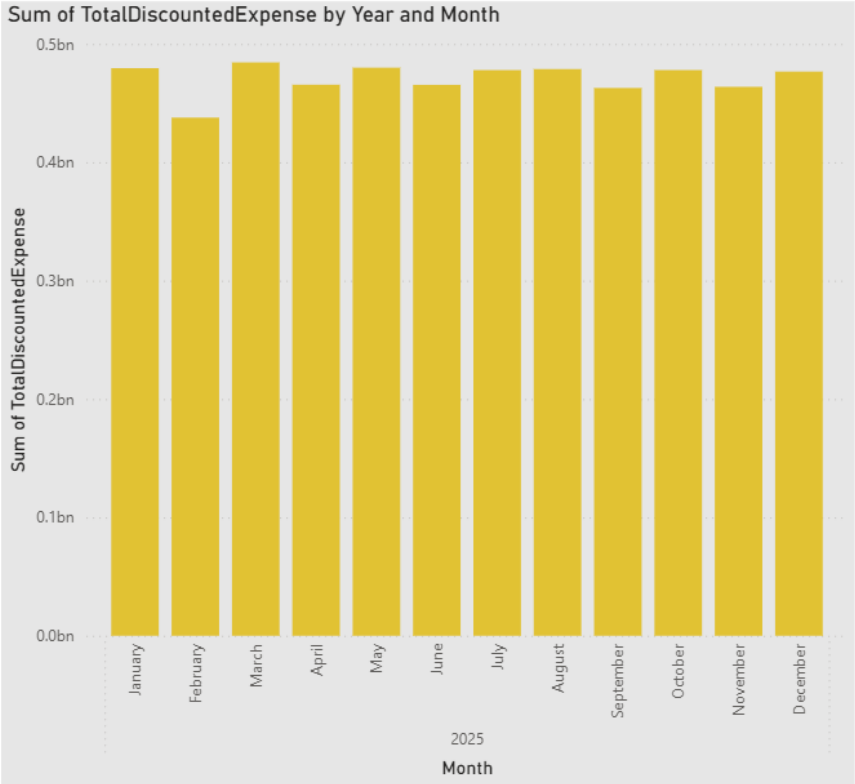
3. Report 3 (Drill-Down Report): Drill-down is used to explore data across multiple levels of a hierarchy. In this project, a Date hierarchy (Year → Month) is used in a bar chart.

Initially, the chart displays the **Total Discounted Expense** by Year. By enabling drill-down, users can click on a specific year to view a more detailed breakdown by month.

This feature helps users move from summarized data to detailed insights, supporting better analytical understanding of trends over time.


Sum of TotalDiscountedExpense by Year
6bn
5bn
4bn
3bn
2bn
1bn
0bn
Sum of TotalDiscountedExpense
2024
2025
2026
Year

atricxes
Page 2
Page 5
Month Slieces
Year Slicies
Drill down


Sum of TotalDiscountedExpense by Year and Month
0.5bn
0.4bn
0.3bn
0.2bn
0.1bn
0.0bn
Sum of TotalDiscountedExpense
January
February
March
April
May
June
July
August
September
October
November
December
2025
Month

atricxes
Page 2
Page 5
Month Slieces
Year Slicies
Drill dc

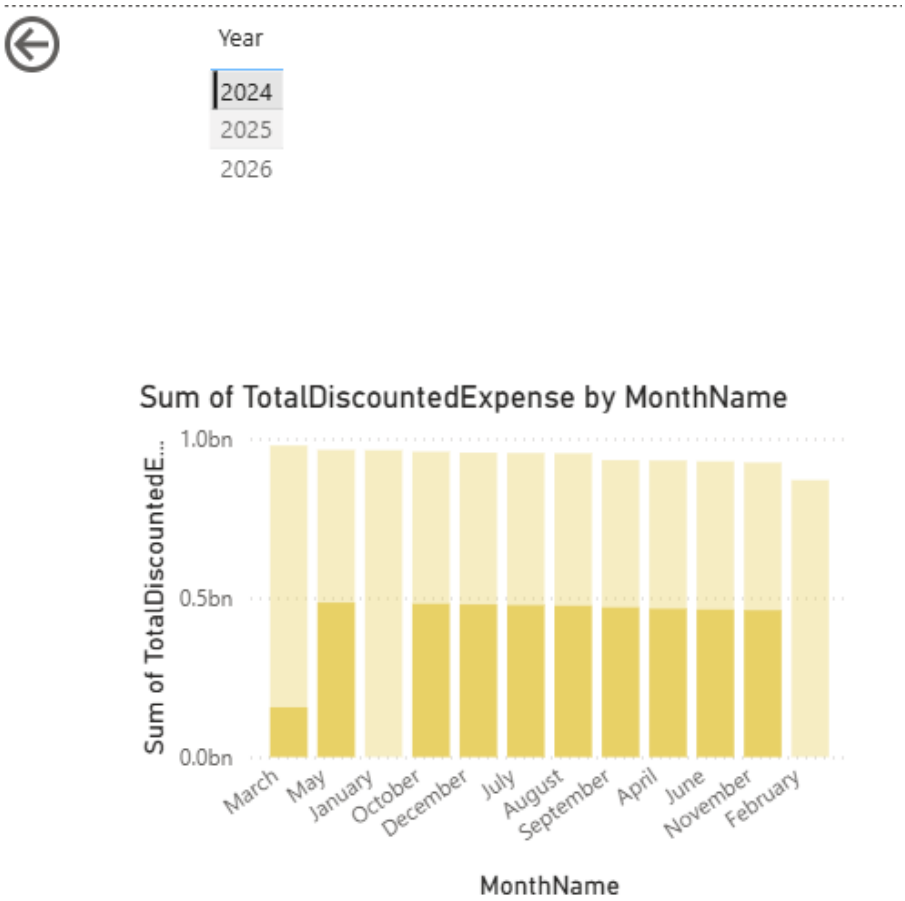
4. Report 4 (Drill-Through Report): Drill-through allows users to navigate to a detailed report page based on a selected data point.

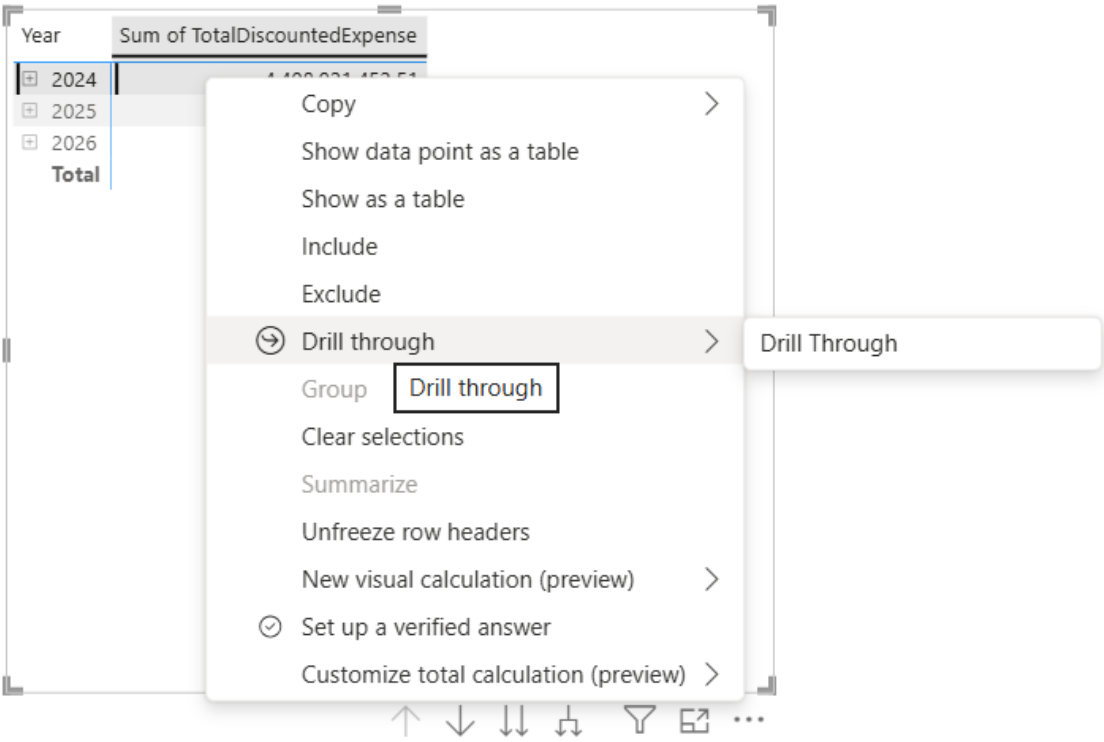
In this project, a drill-through page is created using the Year field. When the user right-clicks on a specific year (e.g., 2024) in a chart and selects *Drill Through*, they are navigated to a detailed page.

This page displays:

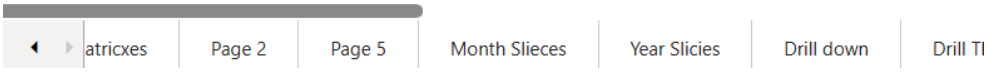
- A Card visual showing total discounted expense
- A Month-wise breakdown chart

The drill-through feature enables deeper analysis of specific data points without cluttering the main dashboard.




Year
Sum of TotalDiscountedExpense
2024
2025
2026
Total
Copy
Show data point as a table
Show as a table
Include
Exclude
Drill through
Group
Drill through
Clear selections
Summarize
Unfreeze row headers
New visual calculation (preview)
Set up a verified answer
Customize total calculation (preview)
Drill Through

11.31bn
Sum of TotalDiscountedExpense


atricxes
Page 2
Page 5
Month Slieces
Year Slicies
Drill down
Drill Tl

5. Conclusion

This assignment demonstrates the creation of a data warehouse, an SSAS cube, and OLAP operations through Excel and Power BI. The use of hierarchies, dimensions, and measures in the SSAS cube ensures that users can perform in-depth analysis on the data. The Power BI reports provide interactive and dynamic views of the data, enabling a detailed exploration of team and match statistics. The documentation includes all necessary steps for each task and shows how the cube and reports were designed and implemented.